General Preparation Methods for Copper Nanorods and Copper Nanowires: A Systematic Review of Solution Combustion Synthesis and Wet-Chemical Routes

Abstract: One-dimensional (1D) copper nanostructures—nanorods (NRs) and nanowires (NWs)—are attractive for flexible electronics, transparent electrodes, and catalysis due to their high aspect ratio, excellent electrical conductivity, and earth abundance. This review systematically examines the preparation methods for Cu NRs/NWs, covering solution combustion synthesis (SCS), hydrothermal synthesis, solution reduction, seed-mediated growth, and template-assisted electrodeposition. The critical roles of the oxidizer-to-fuel (O/F) ratio, capping agents, reducing agents, and reaction temperature in governing phase (metallic Cu versus CuO/Cu_2O) and morphology (nanoparticles versus nanorods/nanowires) are discussed. Based on literature data, typical synthesis parameters and product characteristics are summarized. A key finding is that while SCS with copper nitrate and citric acid readily produces CuO nanorods, direct SCS synthesis of metallic Cu nanowires remains challenging due to the rapid, isotropic combustion process. Wet-chemical routes, particularly hydrothermal and solution reduction methods utilizing alkylamine capping agents, are the predominant approaches for metallic Cu NWs.

Keywords: Copper nanowires; Copper nanorods; Solution combustion synthesis; Hydrothermal synthesis; Capping agents; Oxidizer-to-fuel ratio

1. Introduction

One-dimensional copper nanostructures, including nanorods and nanowires, are important building blocks for next-generation electronics. Copper offers electrical conductivity comparable to silver and gold at a fraction of the cost, and its earth abundance makes it an ideal candidate for commercial-scale applications . However, the high surface reactivity of copper leads to rapid oxidation, and achieving controlled anisotropic growth in one dimension remains a significant synthetic challenge .

The preparation methods for Cu NRs/NWs can be broadly classified into two categories: **combustion-based routes** (including solution combustion synthesis, SCS) and **wet-chemical routes** (including hydrothermal synthesis, solution reduction, seed-mediated growth,

and template-assisted electrodeposition). SCS is renowned for its rapidity, energy efficiency, and molecular-level mixing of reactants, and has been widely applied to metal oxide nanomaterials . However, its application to metallic 1D copper nanostructures faces fundamental difficulties. Wet-chemical routes, by contrast, offer superior control over anisotropic growth through the selective adsorption of capping agents and kinetic regulation of reducing agents.

This review provides a systematic overview of the general preparation methods for Cu NRs/NWs, analyzing the mechanistic principles governing phase and morphology control, and summarizing representative synthesis parameters from the literature.

2. Solution Combustion Synthesis (SCS) Route

2.1 Fundamental Principles and the Challenge for Metallic Copper

Solution combustion synthesis (SCS) is based on an exothermic redox reaction between an oxidizer (typically a metal nitrate) and a fuel (e.g., citric acid, glycine, urea). The process begins from an aqueous solution, ensuring molecular-level mixing, and is triggered by heating, leading to a self-sustained combustion reaction that completes within seconds to minutes . The rapid release of gases (N_2, CO_2, H_2O, NO_x) during combustion inhibits particle growth, favoring nanosized products .

However, applying SCS to **metallic copper** synthesis presents a fundamental challenge. The stoichiometry of the combustion reaction is governed by propellant chemistry, where metal nitrates have negative oxidizing valences and fuels have positive reducing valences . For $Cu(NO_3)_2$, the oxidizing valence is −10; for citric acid ($C_6H_8O_7$), the reducing valence is +18. While the stoichiometric ratio (O/F = 1) can be calculated, the combustion products under these conditions are not metallic copper.

The reaction pathway for $Cu(NO_3)_2$–fuel systems involves initial decomposition to CuO and Cu_2O oxide phases, followed by reduction to metallic Cu by reducing gases (NH_3, H_2) generated from fuel decomposition in the post-combustion zone . This reduction requires **fuel-rich

conditions ($\varphi > 1$)** to generate a sufficiently reducing atmosphere. This mechanism has important implications for 1D growth: the strongly exothermic combustion and rapid gas release promote isotropic growth, favoring spherical or near-spherical nanoparticles rather than anisotropic structures.

2.2 Copper Nitrate–Citric Acid System: CuO Nanorod Formation

In the $Cu(NO_3)_2$–citric acid system, the oxidizer-to-fuel (O/F) ratio exerts decisive control over the product phase. Christy et al. demonstrated that with a stoichiometric composition (O/F = 1), CuO nanorods with a monoclinic crystal structure are obtained . The FESEM and HRTEM analyses revealed rod-like morphologies with widths of 10–20 nm and lengths ranging from 100 to 250 nm . The XRD pattern exhibited characteristic peaks at $2\theta = 32.4°$, 35.4°, 38.6°, 48.7°, 53.4°, 58.2°, 61.4°, 66.1°, 67.9°, 72.3°, and 75.0°, corresponding to the (110), (−110), (111), (−202), (020), (202), (−113), (−311), (113), (311), and (004) Bragg reflections of monoclinic CuO, respectively .

The role of citric acid in promoting nanorod formation has been attributed to its ability to form stable chelate complexes with Cu^{2+} ions, ensuring homogeneous mixing and facilitating the combustion reaction with higher enthalpy release . The heat released during combustion is greater with citric acid than with other fuels, which promotes crystallite growth and complete combustion, yielding a more crystalline product .

Table 1: Representative SCS Parameters for Copper-Based Nanostructures

Precursor System	Fuel	O/F Ratio	Product Phase	Morphology	Reference
$Cu(NO_3)_2$	Citric acid	Stoichiometric	CuO	Nanorods (10–20 nm width)	
$Cu(NO_3)_2$	Urea/Citric acid	Varies	CuO	Nanoparticles	

2.3 Strategies for Metallic Copper Synthesis via SCS

To achieve pure metallic Cu via SCS, researchers have explored modified approaches. Podbolotov et al. developed a microwave-assisted foam preparation method combined with hexamethylenetetramine (HMT) as fuel, enabling one-step synthesis of pure metallic Cu nanoparticles (~67 nm) in air without post-reduction treatment . Trusov et al. employed spray solution combustion synthesis (SSCS) with glycine as fuel ($\varphi = 2$) at 600–750 °C, producing copper hollow microspheres (diameter ~4 μm, wall thickness ~10 nm), though the product was spherical rather than 1D .

Key limitation: The literature contains no reports of direct SCS synthesis of pure metallic Cu nanorods/nanowires using copper nitrate and citric acid. The primary obstacles are: (1) the rapid exothermic combustion and gas release are unfavorable for anisotropic growth; (2) metallic Cu reduction requires fuel-rich conditions, but fuel-rich combustion typically leads to morphological失控 (agglomeration, irregular shapes). Consequently, the predominant routes to Cu NRs/NWs have shifted to wet-chemical liquid-phase methods.

3. Wet-Chemical Routes

3.1 Hydrothermal Synthesis

Hydrothermal synthesis is the most widely used method for Cu nanowire production. The principle relies on the selective adsorption of capping agents (typically alkylamines) on specific crystallographic facets of Cu, directing anisotropic growth. A typical hydrothermal Cu NW synthesis system comprises a copper salt precursor ($CuCl_2$ or $Cu(NO_3)_2$), a reducing agent (glucose, ascorbic acid), a capping agent (hexadecylamine, HDA; or oleylamine, OAm), and an aqueous/organic solvent.

Representative protocol ($CuCl_2$–glucose–HDA system) :

- Copper chloride (2.4 g), D-glucose (3.9 g), and hexadecylamine (14.55 g) are added to 900 mL deionized water

- The mixture is stirred for 12 hours to form a uniform emulsion

- The emulsion is transferred to a hydrothermal reactor and heated for 6–15 hours

- The resulting Cu nanowire solution is collected by centrifugation and washed

Oleylamine–glucose system: Etika et al. synthesized Cu nanowires using $CuCl_2$ as precursor, oleylamine (OAm) as capping agent, and glucose as reducing agent under hydrothermal conditions (120 °C, 7 h, 198 kPa) . The CuP:OAm:dextrose molar ratio significantly influenced the product:

- A **1:3.75:1.1** ratio produced Cu nanowires with an average diameter of 96 nm

- Higher OAm concentrations led to increased nanowire diameters

- The nanowires exhibited a face-centered cubic (FCC) structure

- Electrical conductivity reached up to 1.38×10^5 S cm^{-1}

Other hydrothermal systems: Cu NWs with diameters of 25–30 nm and lengths of 50–65 μm have been synthesized using HDA as the capping agent . The diameter and aspect ratio can be tuned by adjusting reaction time and temperature, with typical products having diameters of 60–100 nm and lengths of 10–90 μm.

3.2 Solution Reduction (Liquid-Phase Reduction)

Solution reduction is conducted at lower temperatures (90–120 °C) using mild reducing agents. The classic method developed by Zeng and co-workers uses $Cu(NO_3)_2$ as precursor, NaOH for pH adjustment, ethylenediamine (EDA) as capping agent, and hydrazine (N_2H_4) as reducing agent at 60 °C, producing Cu nanowires with diameters of 90–120 nm and lengths of 40–50 μm .

Meng and Jin modified this synthesis to improve understanding of the growth mechanism . In a typical reaction, 10 mL of aqueous solution containing 5 mM $Cu(NO_3)_2$, 7 M NaOH, 0.25 M EDA, and 10 mM N_2H_4 (molar ratio 1:1400:50:2) was sealed in a 20 mL glass vial and heated at 60 °C for 40–90 minutes. The solution color turned from royal blue to bronze after

approximately 38 minutes, indicating Cu NW formation. Products were collected by centrifugation, washed with deionized water, and air-dried .

Table 2: Representative Hydrothermal and Solution Reduction Protocols for Cu Nanowires

Method	Precursor	Capping Agent	Reducing Agent	Conditions	Product Dimensions	Reference
Hydrothermal	$CuCl_2$	HDA	D-Glucose	120 °C, 6–15 h	—	
Hydrothermal	$CuCl_2$	Oleylamine	D-Glucose	120 °C, 7 h	~96 nm diameter	
Solution Reduction	$Cu(NO_3)_2$	EDA	N_2H_4	60 °C, 40–90 min	90–120 nm diameter	

A continuous flow reactor (CFR) approach has further improved Cu NW synthesis, yielding longer nanowires under milder chemical conditions . Recent advances in continuous stirred-tank reactor (CSTR) systems have achieved stable operation for 44 hours with a total yield of 71.5%, producing Cu nanowires with lengths of ~45 μm and diameters of ~170 nm .

3.3 Seed-Mediated Growth

Seed-mediated growth uses pre-formed heterogeneous seeds (e.g., Au nanocrystals) to direct the anisotropic growth of Cu, yielding highly monodisperse Cu nanorods. This approach is particularly effective for Cu–Au bimetallic nanorods and can be adapted for Cu-rich structures.

Representative protocol (Au-seeded Cu NRs) :

- **Step 1**: Au nanocrystal seeds are synthesized by reducing $HAuCl_4 \cdot 3H_2O$ with tert-butylamine borane (TBAB) in oleylamine (OLAM) at 15 °C

- **Step 2**: $CuCl_2 \cdot 2H_2O$ (0.5 mmol) is mixed with 10 mL oleylamine in a three-neck flask, degassed, and heated to 80 °C to form a blue Cu(II)-OLAM complex

- **Step 3**: The solution is heated to 180 °C, where Cu(II) is reduced to Cu(I), yielding a yellow solution

- **Step 4**: Au seed solution is injected, and the reaction proceeds at 180 °C for 1 hour

- Products are isolated by isopropanol precipitation and centrifugation

The aspect ratio of the resulting nanorods can be tuned by varying the seed concentration. For Cu_3Au nanorods, decreasing the Au precursor concentration increased the aspect ratio from 1:1 to 4:1, with corresponding length increases from 12.7 ± 2.5 nm to 33.7 ± 4.4 nm . The LSPR peak shifted linearly from 572 to 781 nm with increasing aspect ratio .

Key advantage: Seed-mediated growth provides better batch-to-batch consistency and allows precise control over nanorod length and aspect ratio by adjusting the seed-to-precursor ratio .

3.4 Template-Assisted Electrodeposition

Template-assisted electrodeposition uses porous membranes (e.g., track-etched polycarbonate or anodic aluminum oxide) as templates to confine Cu growth into 1D structures. Copper nanowires are electrodeposited within the cylindrical pores of the membrane, and the template is subsequently dissolved to release the nanowires .

The XRD analysis of electrodeposited Cu nanowires confirmed a face-centered cubic crystal structure, and FESEM revealed continuous, well-aligned nanowires with uniform diameters and high aspect ratios . The optical absorption spectra exhibited a peak at 568 nm attributed to surface plasmon resonance . The diameters are controlled by the pore size of the template, with minimum diameters of 15 nm achievable . The preferred growth direction is [110] due to H^+ ion absorption .

3.5 Mechanism of 1D Growth: Screw Dislocations

A significant mechanistic insight into solution-grown Cu nanowires was provided by Meng and Jin, who demonstrated that the one-dimensional anisotropic growth of Cu NWs and nanotubes in solution is driven by **axial screw dislocations** . Three types of TEM evidence conclusively supported this mechanism: rigorous two-beam TEM analysis identified pure screw dislocations along the ⟨110⟩ direction; twist contour analysis confirmed the presence of Eshelby twist associated with the dislocation; and spontaneously formed hollow nanotubes were observed .

This dislocation-driven growth mechanism contrasts with the classical vapor-liquid-solid (VLS) mechanism and provides a new framework for understanding and controlling Cu NW synthesis. By manipulating supersaturation to promote dislocation-driven growth, Meng and Jin improved NW growth using a continuous flow reactor to yield longer Cu nanowires under milder conditions .

4. Phase Control and Oxidation Resistance

A critical challenge for Cu NRs/NWs is their susceptibility to oxidation. Quantitative XRD analysis of as-prepared Cu nanowires revealed a composition of 98.6% Cu, 0.5% CuO, and 0.9% Cu_2O. After thermal annealing, the composition shifted dramatically to 2.8% Cu, 56.0% CuO, and 41.2% Cu_2O, demonstrating rapid oxidation upon heating . Laser annealing at optimized parameters (100 mW, 1000 nm/s) maintained 95.9% Cu content, indicating that photonic processing can preserve the metallic phase .

Strategies for enhancing oxidation resistance include metallic coating (Ag, Ni), organic molecule modification, and surface passivation . For transparent electrode applications, high-temperature annealing in H_2 atmosphere or vacuum is commonly employed to decompose organic residuals and reduce surface oxides, while simultaneously welding nanowire junctions to reduce contact resistance .

5. Conclusions and Perspectives

The synthesis of Cu nanorods and nanowires has advanced significantly, yet challenges remain. Solution combustion synthesis, while effective for CuO nanorods (10–20 nm width), is not currently viable for direct synthesis of metallic Cu nanowires due to the isotropic nature of the combustion process and the difficulty of maintaining a reducing atmosphere while promoting anisotropic growth. Wet-chemical routes—particularly hydrothermal synthesis with alkylamine capping agents and solution reduction with EDA—are the dominant methods for metallic Cu NWs, achieving diameters of 60–170 nm and lengths of tens of micrometers.

Emerging directions include: (1) continuous flow reactor designs for scalable production ; (2) seed-mediated growth for precise aspect ratio control ; (3) fundamental understanding of dislocation-driven growth mechanisms to rationally design synthesis conditions ; and (4) advanced anti-oxidation strategies to enable practical applications . The rational synthesis of Cu NRs/NWs with controlled size, aspect ratio, and oxidation resistance will continue to drive innovations in flexible electronics, transparent electrodes, and catalysis.

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